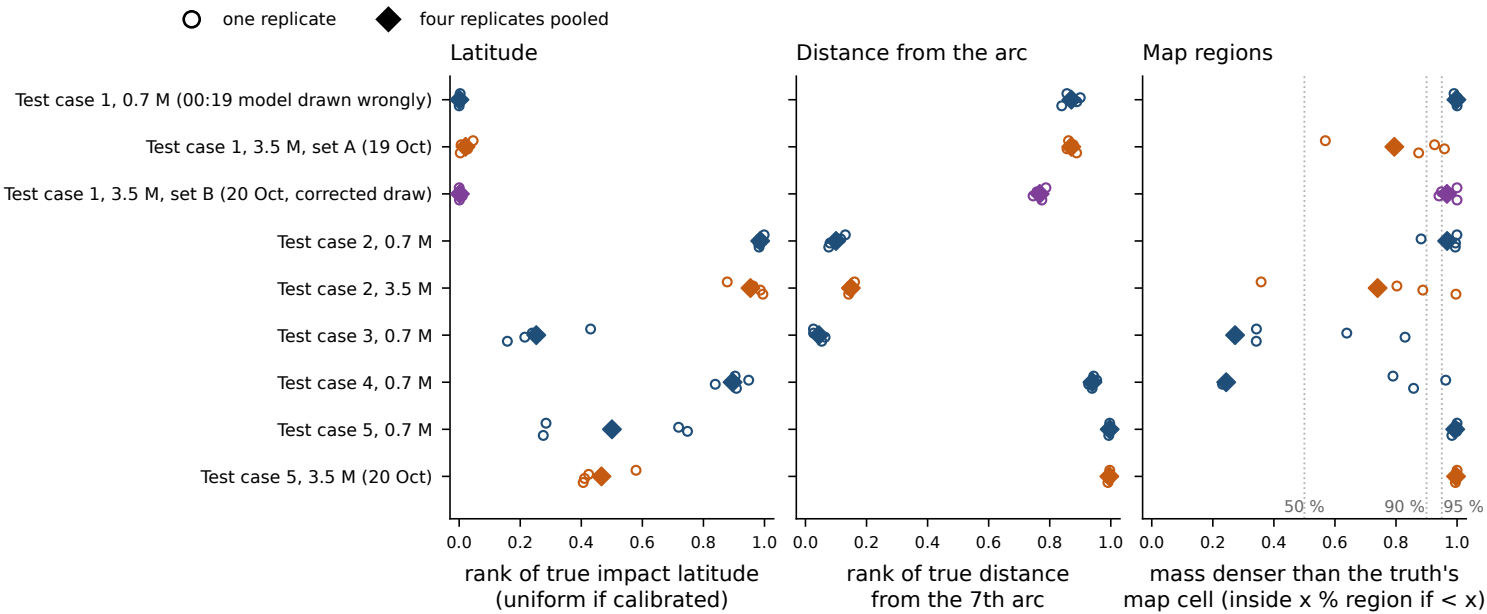
Where the true impact falls in the posterior, per test case and sample size (00:19 data held out)



Note (simplified English): each row is one test flight made from the model. Circles show the four replicate posteriors, the diamond shows them pooled. Test case 1 was run twice at 3.5 million particles with different random numbers (sets A and B, same data before 00:19); the two results do not agree, so the Monte Carlo noise is still large for this case. Test case 5 misses the truth in distance from the arc at both sample sizes with almost no spread: this is not sample size. The cause is in the test generator: it chose the flight at 00:11 without its chance to explain the 00:19 log-on (see note). Physics of the end of flight is provisional.

Technical: statistics per protocol 5 (pre-registered 19 Oct 2026): u = P_post(Q < Q_true); map level = posterior mass in equal-area cells (0.25° × 769 km²) denser than the truth's cell. Core evidence (flight + end of flight), 00:19 option D0 (held out), log-on cause mixture q = 1/3, survival factor; waypoints stratum only; 4 replicates per set; 0.7 M and 3.5 M particles per replicate (1/10 and 1/2 of run J production); 10^4 / 2×10^4 hand-off rows, 64 descents each. Truths from the fuel-only engine prior with the data event BTO(00:11) 17,540–18,540 µs, BFO(00:11) 202–302 Hz. Five cases cannot test coverage: smoke only.